



A Critical Study of Recent Deep Learning-Based Continuous Sign Language Recognition

Hanan A. Taher^{1,2} · Subhi R. M. Zeebaree³

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Abstract

With the growing global emphasis on accessibility and inclusion for deaf and hard-of-hearing individuals, research in Sign Language Recognition (SLR) has gained significant momentum. Sign languages have unique grammar and syntax and rely on manual communication to convey meaning, which presents specific challenges for automated recognition systems. Unlike isolated sign language recognition, continuous sign language recognition (CSLR) must accurately interpret sequences of gestures without clear boundaries between signs. This requires advanced techniques for both segmentation and recognition. This paper presents a comprehensive review of Continuous Sign Language Recognition (CSLR), focusing specifically on deep learning (DL) techniques. It addresses key challenges, such as movement epenthesis (ME)—the transitional movements between signs. Implicit models, including Hidden Markov Models (HMMs) and Connectionist Temporal Classification (CTC), have demonstrated superior performance compared to traditional methods. The researchers reviewed 32 studies published by major publishers (IEEE, Elsevier, and Springer) and examined 12 benchmark datasets related to CSLR. This examination included an overview of their linguistic scope, recording setups, vocabulary size, and participant diversity. Additionally, the performance metrics and methods used in these studies were thoroughly analyzed, followed by a discussion of the results and their broader implications. The review highlights several limitations in existing research, underlining the need for ongoing innovation within the CSLR domain. The insights gained from this review not only enhance the understanding of Sign Language Recognition but also provide a foundation for future research aimed at tackling persistent challenges in this evolving field.

Keywords Sign language · CTC · Deep learning · Movement epenthesis · Gesture recognition

1 Introduction

According to the World Health Organization [1], 466 million people globally—comprising 432 million adults and 34 million children—suffer from hearing loss. Approximately one-third of individuals aged 65 and older experience disabling hearing loss. Those who have minimal or no hearing ability are referred to as “Deaf.” Deaf individuals communicate using sign language, which incorporates facial expressions, body language, and hand movements to express thoughts and emotions, unlike spoken language, which relies on sound. A common misconception is that sign languages are not complete languages; in fact, they have their own grammar and syntax. Sign languages are complete languages that possess unique grammatical structures [2]. They differ from spoken languages in their grammatical organization and they are linguistically independent from each another. There are hundreds of different sign languages, each often tailored to specific regions. For example, American Sign Language (ASL) and British Sign Language (BSL) exist in countries where the same spoken language is prevalent. This variation in sign languages is largely due to their creation by local deaf communities, influenced by geographic location and cultural practices [2].

SLR systems act as intermediaries between the deaf and hearing communities by translating sign language into spoken or written forms. In the last decades, research in SLR has increased significantly due to a global effort to integrate deaf individuals into society. SLR can be categorized into three main types: numbers and alphabets, words, and sentences. The recognition of numbers and alphabets primarily involves static gestures, while words rely mainly on dynamic gestures. Sentences include a combination of both static and dynamic gestures, which must be distinct for accurate translation. The recognition of alphabets and words is called isolated recognition, where one gesture is interpreted at a time. In contrast, the recognition of sentences is known as continuous recognition, as it involves performing multiple gestures sequentially without clear boundaries.

In comparison to isolated sign language recognition (SLR), continuous sign language recognition (CSLR) presents more challenges because videos often do not clearly indicate the breaks between signs. This complexity makes it essential to identify the beginning and end of each sign within a series of consecutive frames [3]. Computer-supported sign language recognition offers significant benefits for deaf and hard-of-hearing individuals by enhancing communication access in various environments, such as education, the workplace, and healthcare settings. Numerous applications powered by artificial intelligence, including robotics, virtual reality games, interpretation services, and automatic video captioning, can leverage CSLR models [4]. Over the past decade, CSLR has attracted considerable attention from researchers, largely due to the rapid advancements made in deep learning (DL). However, despite extensive research efforts, there remains ample room for improvement. Current CSLR systems are still far from achieving their full potential and becoming commercial products, as they struggle with real-time applications and have a limited vocabulary compared to voice recognition systems.

Traditional methods of sign language recognition (SLR) typically use hand-crafted features and rule-based algorithms to interpret signs. However, these methods are limited by their dependence on predefined gestures and their inability to adapt to variations in sign language expressions. In contrast, deep learning approaches utilize neural networks to automatically learn and extract features from large datasets, enabling more accurate and robust recognition of diverse signs. Deep learning models are particularly skilled at handling the complexity and variability of sign language, including subtle differences in gestures and continuous signing. This makes them more effective and scalable than traditional methods [4].

When assessing the performance of CSLR systems, two key metrics are commonly used: Accuracy and Word Error Rate (WER). Accuracy measures the proportion of correctly recognized signs or words out of the total, providing a clear percentage of correct predictions. In contrast, WER offers a more detailed evaluation by considering insertions, deletions, and substitutions in the predicted sequence compared to the ground truth. This makes WER particularly well-suited for tasks involving sequence recognition. While accuracy provides a broad overview of the model's performance, WER offers detailed insights into the errors affecting the sequence and context of signs. Therefore, the combination of both metrics is essential for a comprehensive assessment of CSLR systems [3].

The papers included in this review were selected based on specific criteria to ensure that they represent high-quality and impactful research. We focused on studies published within the last 10 years that specifically address CSLR using deep learning models, such as CNN-HMM, CNN-RNN, 3D CNN, and CNN-TempConv-RNN. Key performance metrics, including accuracy and Word Error Rate (WER), were taken into account, alongside the use of publicly available CSLR benchmark datasets.

Additionally, a critical comparison of older and newer datasets was conducted in this study to provide a comprehensive understanding of how advancements in dataset design have influenced the development of CSLR models. The current study prioritized papers that introduced novel techniques, showed significant improvements, or were highly cited and influential in the field. All selected papers were published in reputable journals. Figure 1 illustrates the distribution of the publications, showing that 75% originated from major publishers such as IEEE, Elsevier, and Springer, while the remaining 25% came from other publishers. The aim of this study is to provide a literature review on CSLR based on deep learning, while also addressing challenges and identifying existing research gaps. The main contributions of this study are summarized as follows:

- Presenting an in-depth analysis of the CSLR task, including its challenges and problem description.
- Presenting an overview of the publicly available benchmark CSLR datasets, including sign language, capture devices, number of words, sentences, signers, the presence of numbers, and alphabets in each dataset.
- Exploring important facets of CSLR (movement epenthesis detectors, and data capture methods).

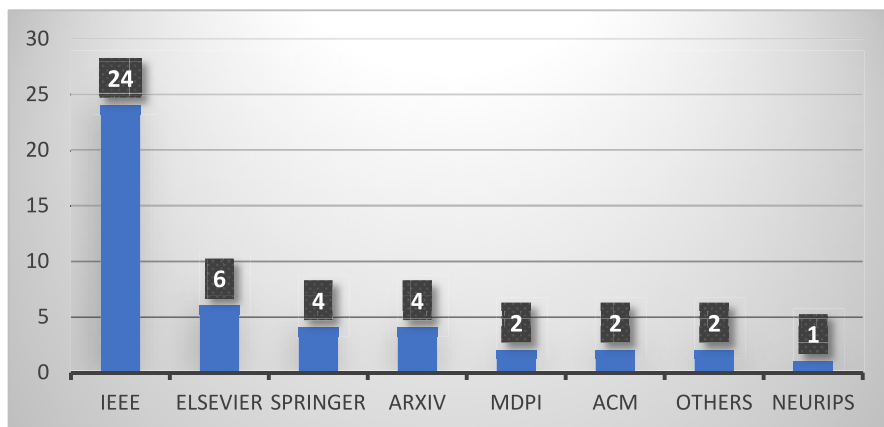


Fig. 1 The distribution of reviewed papers according to publishers

- Literature from the past 10 years is reviewed and investigated based on several criteria.
- Identifying research gaps and limitations within the field of CSLR.

This review is organized as follows: Sect. 2 presents a comprehensive discussion of the challenges associated with CSLR and the current solutions. Section 3 summarizes the various acquisition devices. Section 4 offers a review of CSLR datasets. A detailed analysis of CSLR frameworks is provided in Sect. 5. Section 6 covers the discussion and results. Section 7 identifies research gaps and limitations. Finally, Sect. 8 concludes the study and highlights directions for future research.

2 CSLR Challenges and Current Solutions

CSLR involves identifying and interpreting sign language gestures that are performed seamlessly without pauses. This area of research is particularly significant due to the increasing number of individuals with hearing impairments who rely on sign language for communication. CSLR is a sequence-to-sequence task that aims to convert a series of video frames ($X = \{x_1, x_2, \dots, x_n\}$) into a corresponding series of glosses ($Y = \{y_1, y_2, \dots, y_m\}$). In this context, the number of frames (n) may not match the number of glosses (m), and the glosses must reflect the order of signs in the video. The primary challenge is related to accurately predicting the gloss sequence from a video that depicts a series of signs. This task involves two main difficulties: first, identifying the temporal boundaries of signs in videos with minimal annotations, and second, accurately recognizing the signs themselves. Another significant challenge is the variability among signers. Different signers possess unique physical characteristics, such as skin tone, body type, and height. Additionally, they employ distinct signing techniques influenced by their language proficiency, hand dominance, and signing speed. These variations create substantial obstacles for the

effective generalization of CSLR systems in real-time applications involving new signers that the system has not previously encountered [5, 6].

2.1 Movement Epenthesis (ME) Detection and Solutions

The sentence in sign language consists of a sequence of gestures performed one after another without pauses. Determining the beginning and end of each sign is a challenging problem because the transitions between any two sequential signs are not clearly defined, and the possible hand movements are highly diversified [7]. This phenomenon is known as “movement epenthesis,” which refers to the insertion of transitional movements or non-significant gestures between signs in continuous sign language. These movements are not part of the signs themselves but are necessary for transitioning from one sign to another. Detecting and handling these transitional movements is crucial for accurate SLR systems. Various methods are used to accurately identify and manage these movements (Fig. 2).

1. **Temporal segmentation:** The distinct borders and breaks between signs in CSLR make it challenging to identify the beginning and ending points of individual signs within a continuous statement. To address this issue, several techniques for sign border detection have been devised [8, 9]. One technique is pause-based segmentation, which identifies sign boundaries based on pauses, such as lowering the hands after each sign or pausing briefly between signs [10]. However, this method is often impractical in real-world scenarios [11, 12]. Another approach is

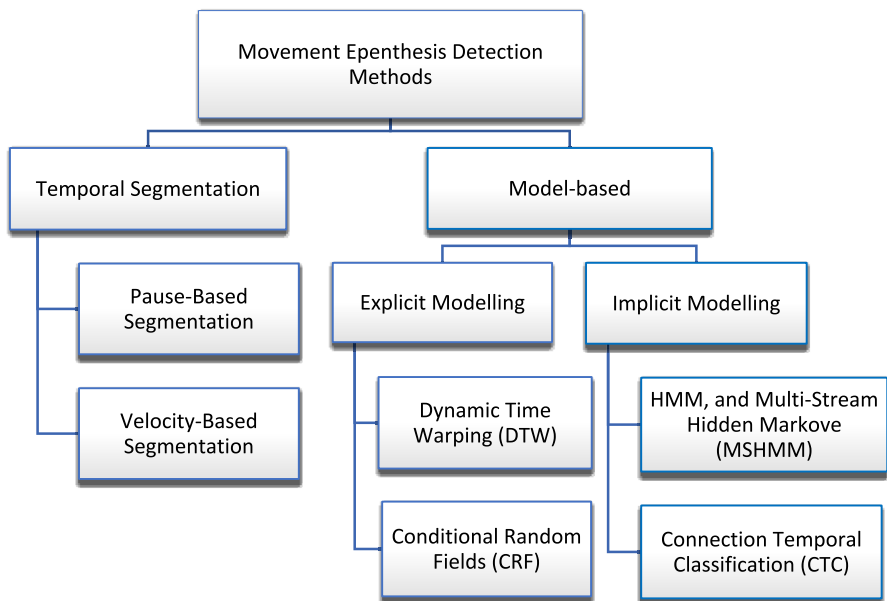


Fig. 2 Various approaches to addressing ME problem

velocity-based segmentation, which relies on changes in hand velocity to segment signs. For example, the zero-crossing of right-hand acceleration can be used to divide sentences into sub-units, which are then analyzed to identify individual signs [13].

2. **Model-based segmentation:** model-based methods are the most popular, trending, and challenging techniques. Providing a more advanced approach to handling the transitions between signs in CSLR. These methods are divided into two main categories:

- *Explicit Modelling:* involves creating specific models to identify and handle ME between signs. Two primary methods used in explicit modeling are Dynamic Time Warping (DTW) and Conditional Random Fields (CRFs). DTW aligns sequences of movements to differentiate actual signs from transitional movements. It calculates the minimum cumulative distance to distinguish between main sign gestures and MEs [14]. CRFs are used for sequence labelling, where each element in a sequence is assigned a label, such as distinguishing between primary signs and MEs. CRFs can also recognize non-gestural movements by assigning specific labels [15]. However, both DTW and CRFs, while effective, can struggle with the high variability and complexity of sign language data. The computational complexity of DTW, highlighted in [16], underscores its high computational cost in both time and space, which can be prohibitive for large-scale datasets or real-time applications.

Scalability Challenges in Explicit Modeling: While DTW and CRFs have proven effective in aligning and managing transitions between signs, their computational complexity can limit their use in large-scale or real-time scenarios. Integrating scalable machine learning algorithms, such as Graph Neural Networks (GNNs) or Sparse Coding, presents an valuable opportunity for improvement. Recent advancements in machine learning have introduced techniques such as GNNs and Sparse Coding, which present promising alternatives to traditional methods like DTW and CRFs for segmentation and recognition tasks in CSLR. GNNs, with their inherent ability to model data as interconnected nodes, can capture both spatial and temporal dependencies in sign language sequences, making them particularly suitable for representing the complex relationships between glosses and transitional movements [17]. For instance, GNNs can model joint movements as graph structures, allowing the algorithm to identify transitions and classify signs effectively. Sparse Coding, on the other hand, focuses on learning parsimonious representations by identifying key features from large datasets while ignoring redundancies, significantly reducing computational load [18]. These methods not only address the high computational demands of DTW and CRFs but also enable scalability for large datasets and real-time applications.

- *Implicit Modelling:* involves more sophisticated methods that do not require explicit segmentation. Two primary techniques used in implicit modelling are Hidden Markov Models (HMM) and Multi-stream HMMs (MSHMMs), and Connectionist Temporal Classification (CTC). HMM utilize hidden states to represent different signs, with observable data (like hand positions and move-

ments) determining the likelihood of these states. MSHMMs enhance recognition by incorporating multiple data streams, such as hand shape, motion trajectories, and facial expressions, thereby improving accuracy and robustness in sign detection [19]. CTC aligns input sequences with target sequences without needing explicit alignment information. It uses a blank label to represent parts of the input that do not correspond to any specific output label, such as unclassified motions in sign language. The CTC loss is calculated by considering all possible alignments between the input and target sequences, summing the losses associated with each alignment to find the best match. This method is effective for real-time processing and handling the complexity and diversity of sign language cues [20, 21].

Implicit modelling techniques, such as HMMs and CTC, are often preferred for CSLR due to their adaptability to signer variability and complexity [22]. These methods eliminate the need for explicit segmentation and efficiently handle transitional movements, enhancing the system's accuracy and robustness in real-world applications [9, 23].

3. **Advancing Hybrid Models for Robustness and Efficiency:** Hybrid modeling approaches that combine explicit and implicit segmentation techniques offer a powerful framework for handling the complexity of CSLR tasks. Explicit models, such as CRFs, excel at fine-grained segmentation by labeling transitions between glosses, while implicit models like HMMs and CTC are adept at aligning sequences without predefined boundaries [24]. By leveraging the strengths of both paradigms, hybrid models could improve both accuracy and efficiency. For example, explicit segmentation could be used to establish preliminary boundaries, providing the implicit model with cleaner inputs for sequence alignment and recognition (51). Such an integrated approach may enhance the generalization capability of CSLR systems, particularly in real-world applications where noise and variability are significant. Recent studies in hybrid modeling for sequence-to-sequence tasks have shown promising results in related domains, further motivating their exploration in CSLR [21].

2.2 Overcoming Occlusion in Depth-Based Hand Tracking

Occlusion in depth-based hand tracking arises when parts of the hands are hidden from the tracking device, hindering accurate movement tracking. Depth-based devices like the Kinect sensor address this by capturing 3D depth information, allowing systems to infer the positions of occluded parts. Depth sensors project infrared light to create a 3D map, aiding detection even when objects are partially hidden [20]. However, challenges persist, particularly with interlocking and crossing hands, where significant occlusion occurs, complicating the tracking of individual fingers and hand movements. These issues are critical in SLR, where precise tracking is essential. Current solutions predict occluded positions from visible data but are not always accurate, especially in complex scenarios, leaving these challenges unresolved.

3 Acquisition Devices

SLR techniques can be categorized as either vision-based [25–27] or sensor-based [28–30], depending on the data acquisition equipment used. In vision-based approaches, a camera captures signs as videos or images. Conversely, sensor-based methods employ devices such as armbands and data gloves to collect and track sign data. These sensor-based methods require less computing power since they do not necessitate image pre-processing. Before 2007, most studies utilized sensors due to the challenges associated with vision-based recognition, which requires extensive measures to capture the necessary features. Sensor-based SLR methods also bypass issues like occlusions, varying illumination, background complexity, and different perspectives common in computer vision recognition. However, their high cost and impracticality limit their real-world application. On the other hand, vision-based approaches are more user-friendly and cost-effective because cameras are widely available. According to the review, most CSLR studies employed a visual approach, with only a few using sensors for sign language acquisition.

3.1 Vision-Based Method

The main input for vision-based CSLR is video data, which is used to record people's sign language movements. To accurately capture the fast movements of sign gestures, high-quality cameras with a high frame rate are required. Since webcams and smartphone cameras are frequently found in homes, they provide a practical means of recording sign language. Modern smartphones come equipped with high-resolution 4K cameras, making them attractive for researchers in sign language capture. Recently, a smartphone's front camera was utilized for capturing sign language gestures [31]. Another popular option is the multi-modality Microsoft Kinect camera, which features an infrared (IR) sensor and two cameras, providing depth, skeleton, and RGB data. The Kinect has frequently been used in research to produce datasets for multi-modal CSLR [32–35].

3.2 Sensor-Based Method

Various sensor-based techniques, such as data gloves, armbands, and smartwatches, have been used to record sign data. Sensor gloves are used in the majority of previous investigations to collect sign data [36–39]. These days, sensor-based CSLR researchers are focusing on less invasive techniques such as using WiFi signals [40].

4 CSLR Benchmark Datasets

An SLR system aims to translate sign language into spoken or written language. For spoken languages, translating between two different languages requires a large vocabulary from both languages. For example, in Google Translate, the

Arabic-English translation system becomes more efficient with more data; it needed over 200 billion words to achieve about 53% accuracy. In contrast, sign language has much smaller vocabulary available. Although there are some benchmark databases for different sign languages, they are limited in size. The largest, RWTH-Phoenix-Weather, has only 1,558 sign words. This lack of comprehensive databases makes current systems impractical and unsuitable for real-time use. Additionally, few efforts have been made in the area of video-based CSLR, primarily due to the lack of appropriate datasets available to train these models. Table 1 highlights the differences in sign languages, capture devices, dataset descriptions (number of words, sentences, and signers for each dataset), and the presence of numbers (123) and alphabets (ABC) among the various sign language benchmark datasets.

The RVL-SLLL American Sign Language (ASL) database [41] has two main parts. The first part features individual video clips that showcase motion primitives and handshapes, including the alphabet, numbers 1–20, various handshapes, and specific signs presented in isolation to illustrate different movements. The second part contains a carefully selected paragraph consisting of two or more sentences that demonstrate connected discourse. This section has been recorded under two different lighting conditions to account for variability in sign recognition. Additionally, the database includes metadata such as signer information, recording conditions, and annotations, which are designed to assist in the analysis and training of CSLR systems. The ASLLVD dataset [42] is a comprehensive collection of ASL videos captured using a multi-camera system. It features 1700 unique signs performed by six different signers, resulting in a total of 2000 video instances. Each video instance is annotated with detailed information, including the start and end frames of each sign, handshapes, and facial expressions, providing a rich resource for studying and recognizing ASL gestures. The dataset also includes variations in signing speed and signer demographics, making it a valuable tool for developing robust and generalized CSLR models. RWTH-BOSTON-104 and RWTH-BOSTON-400 [43] are both video-based ASL databases. However, these datasets differ mainly in size and scope. RWTH-BOSTON-104 contains 104 words, making it suitable for initial research, while RWTH-BOSTON-400 includes 406 words, offering more extensive data for comprehensive model training and evaluation. Both datasets originate from the Boston University ASL corpus and provide sign boundary annotations.

RWTH-PHOENIX-Weather 2014 [44] and SIGNUM Database [45] were created for German Sign Language Recognition (SLR). The RWTH-PHOENIX-Weather 2014 dataset includes 1558 vocabulary words and 6861 sentences, signed by 9 different signers. The SIGNUM Database comprises 780 sentences signed by 20 signers and includes 450 fundamental gestures in its lexicon. Unlike these datasets, the Scene-PHOENIX [46] dataset specifically addresses realistic German CSLR by incorporating various artificial backgrounds into the Phoenix2014 dataset. This approach can be applied to other lab-recorded CSLR datasets to achieve more realistic results. Scene-PHOENIX not only includes RGB video data but also provides depth and skeleton information. This multi-modal approach offers a more extensive feature set for machine learning models. The British Sign Language (BSL) dataset, created using the BSL Corpus, comprises 16 signers and includes 1067 words across 2034 sentences from natural conversational settings. It features transcripts of

Table 1 Publicly accessible CSLR benchmark datasets

Dataset	Year	Sign language	Capture device	Dataset description	123	ABC
RVL-SLLL [41]	2002	American	Standard video camera	14 signers 104 words 600 sentences	1–20	Yes
ASLLVD [42]	2008	American	RGB Camera	6 signers 1700 words 2000 sentences	No	No
RWTH-BOSTON-104 [43]	2008	American	Standard video camera	3 signers 104 words 201 sentences	No	No
RWTH-BOSTON-400 [43]	2008	American	Standard video camera	4 signers 406 words 843 sentences	No	No
RWTH-PHOENIX-Weather 2014 [44]	2014	German	RGB Camera	9 signers 1558 words 6861 sentences	No	No
SIGNUM [45]	2007	German	RGB Camera	20 signers 450 words 780 sentences	No	No
Scene-PHOENIX [46]	2022	German	RGB Camera	9 signers 1558 words 6841 sentences	No	No
British Sign Language [47]	2011	British	RGB Camera	16 signers 1067 words 2034 sentences	No	No
FluentSigners-50 [31]	2022	Russian	RGB Camera	50 signers 278 words 173 sentences	No	No

Table 1 (continued)

Dataset	Year	Sign language	Capture device	Dataset description	123	ABC
Signs World Atlas [48]	2015	Arabic	Canon power Shot A490	10 signers 500 words 30 sentences	1–10	yes
Arabic benchmark database [49]	2015	Arabic	Kinect V2	4 signers 1216 words 531 sentences	Yes	yes
ArabSign [50]	2022	Arabic	Kinect V2	6 signers 95 words 50 sentences	No	No

spontaneous, everyday communication between people using BSL. This dataset captures real-life interactions, reflecting the natural flow, context, and variations in language use, including informal expressions and regional differences. Such comprehensive data is invaluable for developing and evaluating SLR systems in real-world settings [47].

FluentSigners-50 [31] was not generated in a controlled environment, in contrast to the previous datasets. Crowdsourcing was used to collect footage from a variety of sources, including webcams and cell phones, in a variety of environmental conditions. This variation in recording settings can improve the CSLR systems' generalization and results in improved performance when it comes to real-world sign gesture recognition. The Signs World Atlas [48] is an image and video-based database of Arabic Sign Language (ArSL). It features 10 signers performing 500 manual and non-manual signs, including isolated words, continuous sentences, alphabets, numbers, lip movements, and facial expressions. The dataset was captured using a Canon PowerShot A490 digital camera under controlled lighting conditions, marking the first attempt to develop a benchmark database for ArSL.

Another notable ArSL dataset is the Arabic Benchmark Database [49]. It is captured using three different types of sensors: Kinect V2, Leap Motion, and a standard HD camera. It comprises 1216 signs or words, including the alphabet and numeric characters, as well as 531 sentences. Four sign language professionals recorded the entire lexicon in various lighting scenarios, ensuring diversity by including two left-handed and two right-handed signers. The ArabSign dataset [50], also an ArSL resource, was recorded with a Kinect V2 camera and includes 50 sentences. Each sentence is recorded with three distinct types of data: skeleton joint points, color, and depth, which are captured simultaneously. In terms of vocabulary size, the RWTH-PHOENIX-Weather 2014 dataset has the largest vocabulary, while the FluentSigners-50 datasets have the largest number of signers, with 50 signers in each, as outlined in Fig. 3a, b.

Among the available datasets, RWTH-PHOENIX-Weather 2014 and SIGNUM are the most widely utilized. The RWTH-PHOENIX-Weather 2014 dataset consists of recordings of weather forecasts by the German public TV channel Phoenix. Figure 4 displays a frame taken from the original video stream broadcasted

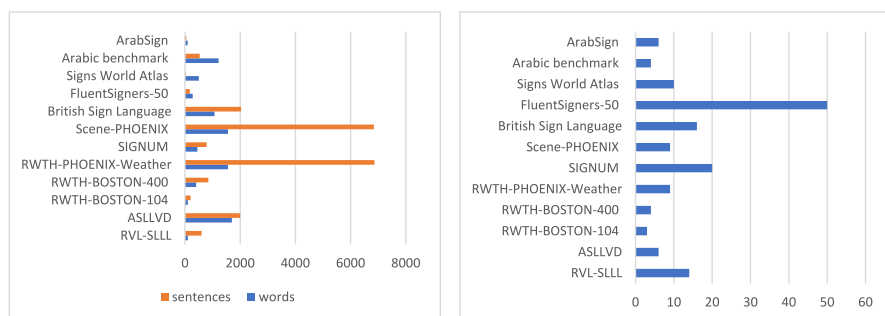


Fig. 3 a, b Statistics of CSL datasets comparison, a words and sentence distribution and b number of signers per dataset



Fig. 4 RWTH-PHOENIX-Weather sample, signer overlaid on the right side

by PHOENIX. In this frame, the weather forecast is overlaid with a sign language interpreter. This overlay results in a spatial resolution of 210×260 pixels. This dataset is particularly challenging due to its real-life context and the complexity of the weather-related content. In contrast, the SIGNUM dataset, developed in a controlled laboratory setting, presents fewer difficulties. The RWTH-PHOENIX-Weather 2014 dataset contains 1558 sign glosses, representing substantial vocabulary. However, this vocabulary size is relatively modest compared to those used in related tasks within Natural Language Processing. Research is ongoing to determine the feasibility of developing commercial CSLR systems with extensive vocabularies that can operate in real time. Figure 5 presents some CSL benchmark datasets samples.

The evolution of publicly available CSLR datasets over the past decade reflects a substantial shift in their complexity and feature richness, directly impacting the development of advanced recognition models. Early datasets such as RWTH-BOSTON-104 and SIGNUM were recorded in controlled environments and featured limited vocabulary, often relying on sensor-based methods or basic RGB data [43]. In contrast, more recent datasets, including RWTH-PHOENIX-Weather 2014 and Scene-PHOENIX, have expanded vocabularies, incorporated diverse signers, and adopted vision-based approaches using multi-modal inputs like RGB, depth, and skeletal data [51]. Notably, Scene-PHOENIX enhances realism by simulating real-world conditions with varying backgrounds, addressing the generalization challenges faced by earlier datasets [52].

These advancements have driven improvements in CSLR models, enabling better handling of challenges such as movement epenthesis and signer variability. For instance, datasets like FluentSigners-50, which utilize crowd-sourced recordings from smartphones and webcams, emphasize diversity in environmental conditions and signer demographics, enhancing model robustness in real-world applications [31]. Moreover, the inclusion of temporal and spatial annotations in newer datasets supports the development of more sophisticated deep learning frameworks that leverage these features for sequence learning and alignment [53]. By comparing older and newer datasets, it is evident that this progression in dataset construction aligns with the broader shift in CSLR technologies toward

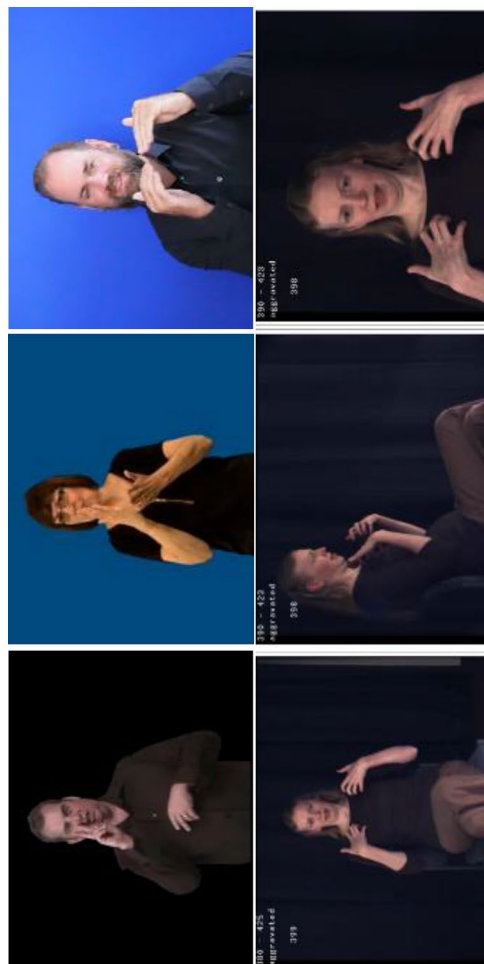


Fig. 5 Samples from CSLR datasets. ASLLVD (Top), BSL (bottom)

real-world applicability and scalability, emphasizing the importance of continued efforts to develop linguistically diverse, multi-modal, and environmentally realistic datasets [23].

5 Impact of Dataset Characteristics on CSLR Model Performance

An analysis of CSLR datasets reveals the significant impact of environmental conditions, signer diversity, and dataset variability on model performance and generalization. Table 2 provides a comparison of key datasets, including RWTH-PHOENIX-Weather (controlled environment), FluentSigners-50 (uncontrolled, real-world environment), CSL (laboratory setting), and Scene-PHOENIX (augmented/realistic environment). Datasets recorded in controlled environments, such as RWTH-PHOENIX-Weather and CSL, generally achieve lower Word Error Rates (WER). For instance, RWTH-PHOENIX-Weather exhibits a WER of approximately 15%, while CSL achieves around 18% [33, 54]. These results are largely attributed to the uniform recording conditions of controlled environments, which reduce noise and variability, allowing models to focus on sign-specific features. The CSL dataset, in particular, demonstrates that controlled settings can mitigate the challenges posed by high signer diversity, as it includes 50 signers but still maintains a relatively low WER.

In contrast, datasets recorded in uncontrolled, real-world settings, such as FluentSigners-50, face challenges stemming from environmental variability, diverse recording devices, and background noise. Despite having the same number of signers (50) as CSL, FluentSigners-50 exhibits a higher WER of approximately 22% [31]. This disparity underscores the significant impact of environmental conditions on model performance. Similarly, Scene-PHOENIX, which augments RWTH-PHOENIX-Weather with synthetic backgrounds to simulate real-world conditions, experiences a WER of around 20%, indicating that even simulated variability introduces challenges for CSLR models [55].

Signer diversity plays a complex role in influencing model generalization. While datasets with a greater number of signers, such as CSL and FluentSigners-50, expose models to a broader range of signing styles, the controlled conditions of CSL allow the model to effectively learn from this diversity, achieving lower WER compared to FluentSigners-50. This suggests that while signer diversity is crucial for generalization, it must be paired with consistent and high-quality data acquisition to achieve optimal results.

It is worth pointing out here that dataset characteristics and recording environments are not the only determinants of CSLR model performance. The training methodology plays an equally critical role. Variations in methodologies can substantially influence performance metrics, complicating efforts to isolate the specific impact of dataset characteristics. To address this, future research should focus on standardizing training and evaluation methodologies across diverse datasets to facilitate clearer and more reliable comparisons, thereby enabling a deeper understanding of the relative contributions of environmental factors, signer diversity, and dataset variability to model performance.

Table 2 Comparative analysis of CSLR datasets and model performance

References	Dataset	Environment	Signers	Challenges	WER (%)
Koller et al.[54]	RWTH-PHOENIX-Weather	Controlled	9	Limited real-world variability	~ 15%
Mukushev et al.[31]	FluentSigners-50	Uncontrolled	50	High environmental noise, diversity	~ 22%
Huang et al.[33]	CSL	Laboratory	50	Lack of real-world diversity	~ 18%
Jang et al.[55]	Scene-PHOENIX	Augmented/Realistic	9	Synthetic variability introduced	~ 20%

6 Deep Learning-Based CSLR

After 2015, the majority of CSLR researchers shifted away from hand-crafted features, opting instead to explore deep learning techniques for feature extraction. Hand-crafted features are manually selected and designed characteristics from raw data used to enhance the performance of machine learning models. Due to the popularity of Convolutional Neural Networks (CNNs), researchers began using deep CNNs to extract frame-level features, which significantly enhanced the performance of CSLR models. These features are then input into various temporal learning models, such as HMMs and Recurrent Neural Networks (RNNs). In the following subsections, CSLR works were categorized and reviewed based on the most commonly used architectures, such as CNN-HMM, CNN-RNN, CNN-TempConv-RNN, and 3D-CNN. A summary of the reviewed deep learning-based CSLR approaches is presented in Table 3. The table provides a comparative overview of various studies, detailing the reference, year, language, dataset, movement epenthesis (ME), method, and word error rate (WER%). The data flow of deep learning-based CSLR approaches is presented in Fig. 6.

In deep learning-based CSLR systems, data flows through several stages. First, the input sequence, consisting of video frames of sign language, is processed. Spatial modelling then extracts features like hand shapes and facial expressions using Convolutional Neural Networks (CNNs). Temporal modelling follows, capturing the sequential nature of gestures through Recurrent Neural Networks (RNNs) or LSTMs. In sequence-to-sequence learning, the system maps the input sequence to an output, using models like Transformers or attention mechanisms. Finally, the target sequence, such as text or speech, is generated, representing the recognized signs from the input.

6.1 CNN and HMM

Researchers initially favored HMMs for CSLR due to their past success in modelling sign sequences. However, with the rise of neural networks, which can learn features directly from data, there has been a shift away from hand-crafted features towards those learned by neural networks. Neural networks have been widely adopted in CSLR for feature extraction, with HMMs subsequently used to model and identify sign sequences. Studies [56, 57], and [80] demonstrated that incorporating RGB and skeletal data, which provide detailed information about the signers' joints and body poses, helped overcome challenges such as occlusions and improved CSLR performance. For example, Study [56] used a multi-layer perceptron (MLP) for feature extraction and then applied a Gaussian HMM (GHMM) to determine the sign labels. Study [57] modelled cross-language subunits based on hand orientations using HamNoSys annotations. Study [80] developed a multi-stream CNN-LSTM-HMM model, which required annotations for gloss, hand, and mouth shapes to be trained effectively.

Table 3 Summary of the DL-based methods examined for CSLR

Refs.	Year	Language	Dataset	ME	Method	WER (%)
Gweth et al. [56]	2012	Germany	SIGNUM	–	MLP-HMM	16.0
Koller et al. [57]	2016	Germany	Phoenix2014	HMM	CNN-HMM	45.1
Camgoz et al. [51]	2017	Germany	Phoenix2014	CTC	CNN-BLSTM	40.7
Koller et al. [58]	2017	Germany	SIGNUM	CTC	CNN-BLSTM-HMM	4.8
Cui et al. [59]	2017	Germany	Phoenix2014	CTC	CNN-BLSTM	38.7
Pu et al. [60]	2020	Germany	Phoenix2014	CTC	CNN-BLSTM	34.3
Hu et al. [61]	2024	Germany	Phoenix2014	CTC	CNN-1DCNN-BLSTM	20.0
Cui et al. [62]	2019	Germany	Phoenix2014	CTC	CNN-BLSTM	22.8
Hao et al. [63]	2021	Germany	Phoenix2014	CTC	CNN-1DCNN-BLSTM	21.0
Hu et al. [64]	2022	Germany	Phoenix2014	CTC	CNN-1DCNN-BLSTM	20.8
Hu et al. [52]	2023	Germany	Phoenix2014	CTC	CNN-1DCNN-BLSTM	19.4
Jang et al. [65]	2023	Germany	Phoenix2014	CTC	CNN-1DCNN-BLSTM	20.8
Min et al. [66]	2022	Germany	Phoenix2014	CTC	2DCNN-1DCNN-BLSTM	32.2
Zhou et al. [67]	2021	Germany	Phoenix2014	CTC	CNN-TempConv-BLSTM	20.7
Zhu et al. [68]	2023	Germany	Phoenix2014	CTC	CNN-1DCNN-Trans-former	24.7
Pu et al. [69]	2018	Germany	Phoenix2014	CTC	3DCNN-TempConv	37.3
Yang et al. [70]	2019	Germany	Phoenix2014	CTC	2D3DCNN-LSTM-BLSTM	34.9
Chen et al. [71]	2022	Germany	Phoenix2014	CTC	3DCNN	18.8
Huang et al. [33]	2018	Germany	Phoenix2014	CTC	3DCNN-Atten-DTW	38.3
Min et al. [53]	2021	Chinese	CSL Split I	CTC	VAC CNN-1DCNN-BLSTM	1.6
Papastratis et al. [72]	2020	Chinese	Phoenix2014	CTC	CNN-1DCNN-BLSTM	24.0
Hu et al. [73]	2023	Chinese	CSL Split I	CTC	CNN-1DCNN-BLSTM	20.7
Pu et al. [74]	2019	Chinese	CSL Split II	CTC	3DCNN-BLSTM-Atten	32.7
Zhou et al. [75]	2019	Chinese	CSL Split I	CTC	3DCNN-GRU	4.7
Huang and Ye [25]	2021	Chinese	CSL Split I	CTC	3DCNN-BLSTM-Atten	7.4
Hu et al. [76]	2023	Greek	GrSL-SD	CTC	CNN-1DCNN-BLSTM	33.5
Adaloglou et al. [77]	2021	Greek	GrSL-SI	CTC	I3D-BLSTM	2.2
Zhang et al. [40]	2020	American	Private	CTC	DBN-HMM	17.0
Fang et al. [78]	2017	American	Private	CTC	BLSTM	16.1
Sharma et al. [79]	2021	Indian	Private	CTC	CNN-BLSTM	12.4
Suri and Gupta [30]	2019	Indian	Private	CTC	CapsNet	6.0
Brock et al. [80]	2020	Japanese	Private	CTC	CNN-RNN	11.3

6.2 CNN and RNN

RNNs have become more popular in CSLR due to HMMs' inability to fully capture the global context of sign language sentences. RNNs gave CSLR new powers because they were made to manage lengthy dependencies in sequences. An RNN type called Long Short-Term Memory (LSTM) was developed to solve

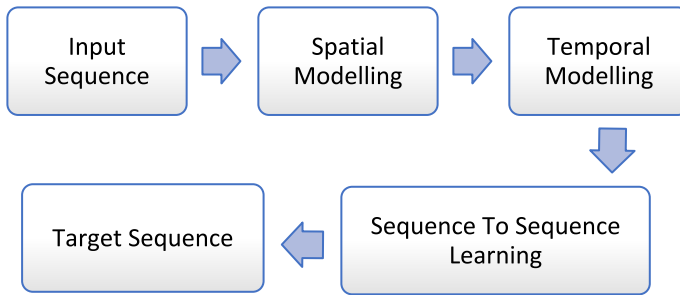


Fig. 6 Data flow in DL-based CSLR systems

problems such as the vanishing gradient problem that are frequently encountered in conventional RNNs. Adoption of bidirectional RNNs, in particular bidirectional LSTMs (BLSTMs), which allowed data to flow from both past and future observations in the sequence, was one important improvement. This bidirectional information flow enabled the learning of increasingly intricate relationships in CSLR.

The first applications of BLSTM for CSLR were found in [51] and [58], where a CNN-BLSTM model was proposed. [51] presented SubUNets, a system that combines CNNs with BLSTMs and CTC. Their work demonstrated the effectiveness of combining two SubUNets that were trained on both cropped hand images and full-frame images. To address the shortcomings in feature extraction, an iterative training strategy was proposed by [58, 59]. This strategy involved training the network using a method similar to expectation maximization and utilizing the network's pseudo labels to fine-tune the feature extractor, resulting in the ReSign model. In comparison to their previous model, Deep-Sign [57], the resulting model, ReSign [58], dramatically lowered the WER by combining CNN-BLSTM with HMM.

To create pseudo pairs of text and video, the authors [60] developed a Cross Modal Augmentation (CMA) method that involves adding, replacing, and deleting text labels from the matching video frames. Nevertheless, the pseudo labels have to be created manually using this procedure. Later, Prior Aware CMA (PACMA) [61], which suggested pseudo labels using a language model, was added to the framework for improvement. To enhance feature extraction, [62] proposed using gloss-level alignment proposals to train the feature extractor. On the Phoenix2014 dataset, this approach reduced the WER by 4% compared to the ReSign model by [58]. CNN-RNNs have been used to model sensor data obtained from the Leap Motion Controller (LMC) [78] and sensor gloves [79], in addition to vision-based techniques. To represent skeletal data, [78] proposed a hierarchical bidirectional RNN (HB-BRNN), and [79] investigated pre-training CSLR models using isolated SLR data. However, this methodology was only examined on a limited private dataset including 40 Indian Sign Language (ISL) sentences.

6.3 CNN, Temporal Convolutions and RNN

Temporal Convolutions (TempConvs) are a method used to analyze sequences of data over time, such as a series of events occurring sequentially. They utilize a type of neural network called a 1D Convolutional Neural Network (1D CNN) and a technique called dilation to capture patterns in this data. Dilation introduces gaps between the filter elements, allowing the model to capture long-range dependencies without increasing the number of parameters. TempConvs was first introduced by [59], combining TempConvs with a pre-trained VGG model (a type of neural network originally designed for image recognition) and a BLSTM layer (a type of neural network that can process sequences in both directions). This combination was used to better understand sequences in sign language, achieving a Word Error Rate (WER) of 38.7% on a dataset called Phoenix2014. However, this approach had a limitation: it only used images of the signer's right hand, ignoring the fact that sign language often involves both hands.

Later research [52, 53, 62–67, 72] expanded on this idea, using a combination of 2D CNN and 1D CNN to capture both temporal and spatial information. Ref. [72] proposed processing gloss sequences using an RNN language model in conjunction with a 2D CNN-TempConv-BLSTM for video frames. Ref. [53] introduced Visual Alignment Constraint (VAC) loss as a new method to reduce overfitting and cut down training time compared to previous optimization methods. Additionally, [66] proposed Radial CTC to address the peaky behaviour of CTC, leading to further performance enhancements.

Efforts to enhance VAC involved implementing Self-Mutual Knowledge Distillation (SMKD) [63], which enabled simultaneous training of the visual and temporal modules. Additionally, Temporal Lift Pooling (TLP) [64] was introduced to replace the max pooling layers in VAC. This change resulted in a 2% improvement in the WER on the Phoenix2014 dataset. Enhancing spatial attention on significant regions inside the frames was the goal of several investigations [52, 65]. Key regions were highlighted in attention maps created by the Self-Emphasizing Network (SEN) [52], while Divide and Focus Convolution (DFConv) [65] analyzed the top and bottom portions of the frame independently to ensure the model focused on both manual and non-manual signals.

Correlation Network (CorrNet) [52] created correlation maps of body movements between consecutive frames, achieving state-of-the-art WERs across multiple datasets using only RGB data. On the other hand, Spatial–Temporal Multi-Cue (STMC) [67] used a multi-modal approach that combined various features (pose, face, hands, and full frame) and trained the framework for both CSLR and translation tasks. However, the multi-modal STMC was less effective compared to the RGB-based CorrNet.

Ref. [76] explored the joint training of CSLR frameworks for multiple sign languages, using distinct pathways for each language and a common visual backbone. This approach resulted in a limited improvement, achieving a 20.9% WER on the Phoenix2014 dataset, despite training on the GrSL-SD, Phoenix2014, and CSL datasets. Recent efforts have focused on optimizing model efficiency without significantly reducing performance. For example, AdaBrowse [73] and Temporal Super Resolution

Network (TSRNet) [68] demonstrated the potential of dynamic resolution adjustments for real-time applications.

6.4 3DCNN

With their demonstrated ability to extract spatio-temporal information, 3D Convolutional Neural Networks (3DCNNs) are an excellent choice for applications such as CSLR. For instance, [69] introduced a 3DResNet-TempConv model trained using an iterative optimization methodology. This framework was later enhanced by incorporating an attention window, resulting in a reduction in WER by approximately 1% [74]. Furthermore, [75] employed a 3DCNN-BGRU with iterative training and pseudo labels, utilizing an I3D model instead of 3DResNet, which led to a notable 6% decrease in WER compared to [74].

In another significant advancement, [70] presented the Structured Feature Network (SF-Net), which leveraged 2D/3D convolutions, LSTM, and BLSTM to extract features at three levels: frame, gloss, and sentence. However, as noted by [25], the performance of CSLR models drastically decreases with long sequences. To address this, they proposed the Boundary-Adaptive Encoder (BAE), which significantly improved performance on the CSL dataset. Additionally, [77] conducted a comparative analysis of several well-known CSLR models, including 3DResNet-BLSTM and I3D-BLSTM [74]. These models were evaluated using three datasets: Phoenix2014, CSL, and a newly released dataset for Greek Sign Language (GrSL). The study found that pre-training CSLR models with a separate dataset of sign language enhances recognition performance.

Refs. [71] and [33] explored multi-stream 3D Convolutional Neural Network (3DCNN) architectures. Ref. [33] introduced a two-stream 3DCNN (C3D) that utilized both hand and full-frame images, combined with a Hierarchical Attention Network and Latent Space (LS-HAN). More recently, [71] presented a two-stream network that independently encodes full-frame images and position heat maps using an S3D model. This method produced state-of-the-art results on multiple datasets.

CSLR literature includes some less common methods, such as Capsule Network (CapsNet) [30] and Deep Belief Network (DBN) [40]. Both studies explored sensor-based approaches. Ref. [30] proposed using CapsNet instead of CNN for CSLR, achieving approximately 7% higher accuracy than CNN. However, their model was evaluated using only twenty Indian Sign Language (ISL) sentences. Ref. [40] utilized WiFi data for CSLR retrieval. To extract features from WiFi data that captured the signer's mobility, they introduced a Deep Belief Network (DBN). This graphical generative model demonstrated strong recognition performance, especially with non-visual material. Subsequently, a Hidden Markov Model (HMM) was employed with the extracted features for temporal learning and classification. After testing 100 ASL sentences, this method achieved an accuracy of 83%.

7 Discussion and Results

In terms of sign language, despite the existence of hundreds of sign languages worldwide, researchers have focused on only a few for CSLR studies. This limited focus is largely due to the lack of publicly accessible datasets for many sign languages. Additionally, the number of users of specific sign language significantly influences researchers' decisions to focus on that language. For instance, German Sign Language, Chinese Sign Language, and American Sign Language are prominent targets. As illustrated in Fig. 7, German Sign Language is the most frequently addressed, featuring in 60% of the studies, largely due to the popularity of the Phoenix2014 dataset, one of the earliest publicly available CSLR datasets. Chinese Sign Language follows, being the focus of 19% of the studies, while other sign languages have been studied only a few times.

In terms of data acquisition used, research done before 2005 mostly used wearable sensor-based methods. Researchers shifted their attention to non-intrusive cameras after 2005. This change was made to produce recordings that were more lifelike. Slight changes in camera orientation and position were added to several recently released datasets, such as the GrSL dataset [77], in order to increase data diversity and improve CSLR system generalization. Furthermore, it is noteworthy that the only CSLR dataset captured with smartphones and tablets is the Fluent-Signers-50 dataset [31]. IT has been predicted that advanced cameras and sensors will give way to more widely available data-capturing devices like smartphones, tablets, and webcams.

In terms of recognition methods, most deep learning-based CSLR models have employed CNNs for spatial feature extraction, as illustrated in Table 2. Researchers have utilized various pretrained CNN backbones, including CaffeNet [51], ResNet [53, 63, 64, 68, 70], GoogLeNet [58, 62], VGG [59], and BN-Inception [72]. In several investigations, 3DCNNs were employed for both spatial and temporal encoding. A variety of pretrained 3DCNNs were used, including 3DResNets [40, 69, 74], I3D model [77], C3D [33], and S3D [71]. Most of these 3DCNNs were pretrained on action recognition datasets such as Kinetics-400, Sports1M, and UCF101 due to the similarities between SLR and action recognition tasks. However, some studies pretrained their 3DCNNs on isolated SLR datasets.

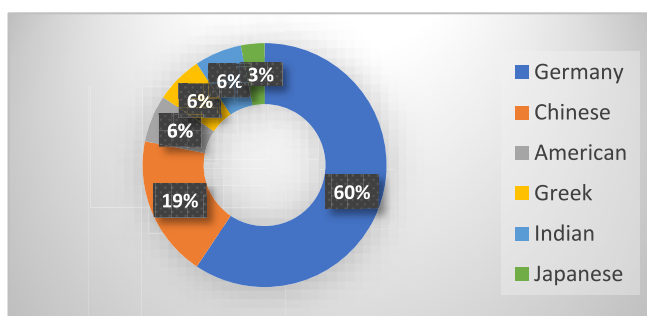


Fig. 7 Distribution of CSLR studies per sign language

In the context of temporal modelling, BLSTM networks have been predominantly employed in previous research to learn time-wise features. The prevalent use of BLSTM can be attributed to its capability to leverage both past and future contextual information simultaneously. This bidirectional processing enhances the network's ability to capture temporal dependencies more effectively, thereby improving the generation and recognition of sign language sentences. The capacity of BLSTM to utilize information from both directions within a sequence makes it particularly well-suited for tasks that require understanding the full context of time-series data. Numerous studies have empirically demonstrated that BLSTM outperforms unidirectional LSTM in CSLR [58, 59] as shown in Fig. 8.

Concerning sequence alignment techniques, nearly all deep learning-based CSLR methods utilize CTC to match frames with gloss sequences. Only a limited number of DL-based CSLR studies have investigated alternatives to CTC, such as HMM. In 2014, the HMM baseline achieved 53% WER using hand-crafted features [54]. The following year, by using CNN-learned features instead of traditional ones in the DeepSign network, a significant WER reduction of 8% was realized [57]. Furthermore, employing RNN and iterative re-alignment in the CNN-BLSTM-HMM model (ReSign) [58] led to a substantial WER decrease of 12%. Researchers then moved away from HMM-based models, adopting a BLSTM-CTC architecture, which improved performance by an additional 7% with iterative training [59].

Ref. [77] led the 3DCNN category with their I3D-BLSTM model on the GrSL-SI dataset, achieving an impressive WER of 2.2%. Their approach demonstrated the effectiveness of 3D convolutional neural networks in capturing spatiotemporal features for accurate SLR. Overall, the best performance across all methods in CSLR is achieved by [53]. Their VAC CNN-1DCNN-BLSTM model, applied to the CSL Split I dataset, achieved an extraordinarily low WER of 1.6%, as shown in Fig. 9. This impressive result highlights the effectiveness of combining CNNs, 1D convolutions, and BLSTMs for capturing both spatial and temporal features, leading to highly accurate SLR.

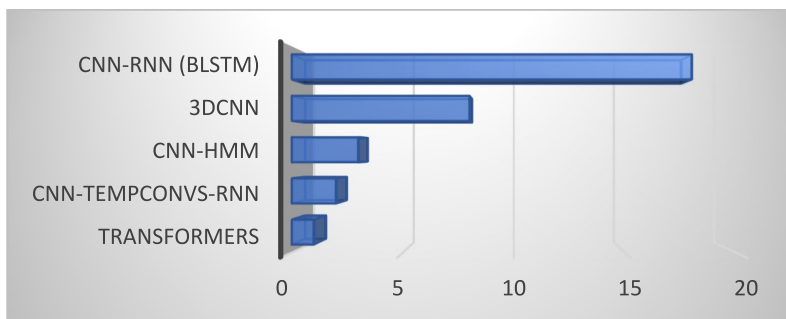


Fig. 8 Percentage of studies per method

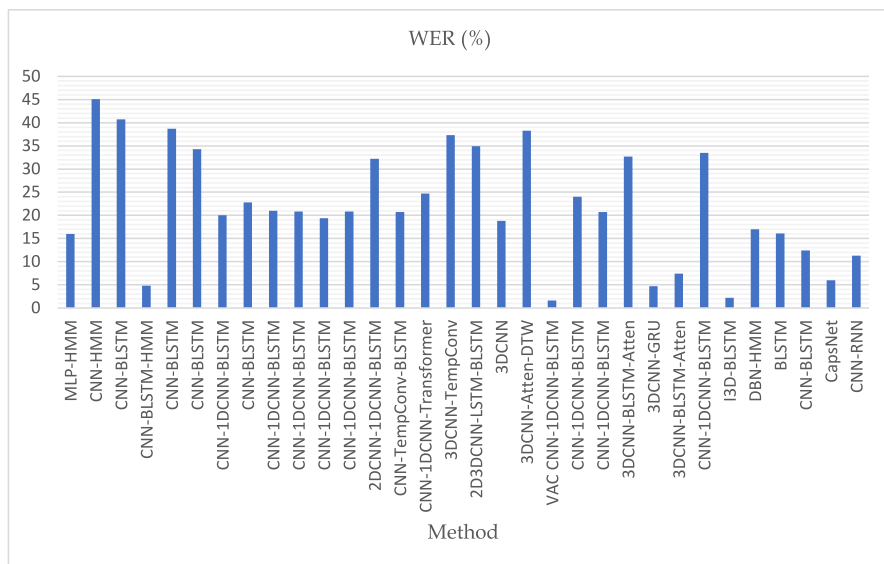


Fig. 9 Comparison based on method and WER

8 Limitations and Research Gaps

- **Optimizing CSLR Training with Limited Data:** Hiring skilled sign language signers to create CSLR datasets is difficult and costly. As a result, generating a significant number of examples is frequently challenging and expensive. Therefore, it is essential to develop CSLR frameworks capable of operating with minimal training data. New methods, such as few-shot learning, can be applied to achieve this. Additionally, a large amount of unlabelled sign language video is available on public websites such as YouTube, which can be leveraged using semi-supervised learning approaches.
- **Signer-Independence:** The performance of CSLR models significantly drops when encountering new signers, primarily due to the limited diversity of signers in existing datasets. One solution is to increase the number of signers in the training set, although this is not always feasible. Alternatively, effective data augmentation methods can be explored to simulate various appearances and signing speeds.
- **Improving CSLR with Multi-Modal Data Integration:** Accurate recognition can be improved by merging input from several modalities, such as text, RGB video, depth, and skeletal information. Nevertheless, multi-modal CSLR systems are computationally demanding, and further study is needed to create effective ways to combine and leverage these various data sources without significantly increasing computational requirements.
- **Lack of CSLR datasets:** Currently, publicly available CSLR datasets are limited and primarily cover only a few sign languages, resulting in the neglect of numerous others. Consequently, significant efforts should be directed towards develop-

ing CSLR datasets for these underrepresented sign languages. This would facilitate a better understanding of their unique features and enable the creation of CSLR models tailored to their specific requirements.

- **Restricted Range of Sign Representations:** Most CSLR datasets encompass only a narrow range of signs. For instance, the Phoenix2014 dataset, which serves as the standard benchmark, includes just 1389 unique signs and is confined to weather forecasts. Consequently, the current state-of-the-art models are designed to operate within a restricted scope and are not yet suitable for commercial applications. To make CSLR models viable for real-world use, it is essential to develop linguistically diverse CSLR datasets that cover a broad array of signs across various domains and scenarios.
- **Selfie-Captured CSLR Datasets:** While CSLR models are anticipated to be used on smart devices like mobile phones and tablets, most existing CSLR datasets were not captured in selfie view with smartphones. Therefore, it is necessary to develop CSLR datasets recorded from a close front selfie perspective using various smartphones and tablets.
- **Multi-person CSLR:** Previous works in CSLR presuppose the presence of a single individual within the frame. Consequently, these models would be ineffective when faced with scenarios involving multiple signers in the same scene. Therefore, there is a clear necessity for CSLR models capable of handling multi-person conversations.
- **Synthetic Data Generation:** The scarcity and variability of high-quality datasets remain a significant challenge in CSLR. Synthetic data generation has emerged as a promising solution to address these limitations. Techniques such as Generative Adversarial Networks (GANs), data augmentation, and simulation-based methods offer potential pathways to enhance dataset diversity and size. For instance, GANs can be trained on existing CSLR datasets to generate realistic video sequences, simulating signs that are underrepresented in the available data. Similarly, augmentation techniques such as flipping, cropping, and scaling can artificially increase variability, helping models learn to generalize better across contexts.

9 Conclusion

CSLR has recently garnered increased attention due to the rapid advancements in machine learning techniques. This paper provides a comprehensive review of CSLR research efforts over the past 10 years. It aims at identifying existing research gaps and promoting the advancement of the CSLR field. The study discusses various challenges associated with CSLR, including sign language, sign boundary detection, data acquisition, and recognition methods. Additionally, it reviews publicly available CSLR datasets in terms of sign language, capture device, number of signs, sentences, and signers. A critical analysis of 32 studies is presented and organized into a taxonomy of five dimensions: sign language, datasets, ME, recognition techniques, and overall performance.

One key finding of the study is that there are very few annotated datasets for CSLR, and these datasets mostly cover only a few sign languages like American Sign Language (ASL), German Sign Language (DSL), and Chinese Sign Language (CSL). This scarcity of annotated datasets explains why there is not much research on CSLR for other sign languages. Most publicly available datasets for CSLR are recorded in controlled environments with limited signs and specific domains, restricting the robustness of CSLR models in real-world scenarios. Additionally, CSLR techniques are shifting from sensor-based to vision-based approaches, using full-frame images for a global representation of features to enhance model accuracy and effectiveness.

Recent CSLR systems use advanced methods for recognizing signs. They often start with pre-trained CNN to identify spatial features from images. After that, TempConvs and BLSTMs are used to understand and learn the sequence of movements over time. Most deep learning models in CSLR use CTC loss to align frames with sign labels. However, because CTC can lead to overfitting, researchers are developing more effective training techniques, such as iterative training plans and additional loss functions, to enhance model performance. Enhancing CSLR using multi-modal data, selfie-captured CSLR datasets, and multi-person scenes in CSLR models are some of the other research gaps and opportunities explored in this study.

Authors Contributions Hanan A. Taher—Conceptualization and Review Design: Lead the design and structure of the study. Data Comparison Tables: Responsible for creating and compiling the two comparison tables, including dataset descriptions and performance metrics. Manuscript Writing: Draft the introduction, methodology, key findings, and discussion sections. Coordination: Handle all communications with the journal, including submission and responses to reviewer comments. Subhi R. M. Zeebaree—Technical Review: Contribute to reviewing deep learning techniques like CNN–HMM, CNN–BLSTM, and CTC in CSLR. Methodology Writing: Assist with writing the sections that focus on model comparisons and performance evaluations. Editing and Proofreading: Provide feedback on the manuscript and ensure technical accuracy.

Data availability No new data was generated or analyzed in this study.

Declarations

Conflict of Interest The authors declare no conflicts of interest.

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Authors and Affiliations

Hanan A. Taher^{1,2}  · Subhi R. M. Zeebaree³

✉ Hanan A. Taher
hanan.taher@dpu.edu.krd; hanan.abdulrahman@auas.edu.krd
Subhi R. M. Zeebaree
subhi.rafeeq@dpu.edu.krd

- ¹ ITM Department, Technical College of Administration, Duhok Polytechnic University, Duhok, Iraq
- ² IT Department, College of Informatics, Akré University of Applied Science, Duhok, Iraq
- ³ Energy Engineering Department, Technical College of Engineering, Duhok Polytechnic University, Duhok, Iraq

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